

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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Platform	ServiceNow Flow Designer
Date	02 October 2026

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1. Introduction

1.1 Project Overview

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer is an automation solution built on the ServiceNow platform using Flow Designer. In many organisations, ordering a laptop for an employee involves repeated manual exchanges between the requester, the IT team and procurement: which specification is needed for the job role, whether it fits the budget, which software must be installed, and whether the request is even standard.

The solution captures the employee's job role and requirements through a catalog request, validates them, compares them with the approved standard laptop configurations (CPU, RAM, storage, GPU, required software and budget), recommends the matching configuration and generates the procurement request automatically. Any requirement that departs from the standard is detected and routed through an exception-handling and approval workflow, so that people spend their time only on the cases that genuinely need a decision.

1.2 Purpose

The purpose of the project is to replace a manual, inconsistent laptop ordering process with a guided and automated one. The goals are to:

- reduce the manual effort IT and procurement staff spend on routine laptop orders;
- ensure every laptop is ordered from an approved standard configuration for the employee's job role;
- validate requirements and budget before a procurement request is raised, reducing rework;
- detect non-standard requirements automatically and handle them through a controlled approval workflow;
- improve consistency, speed and traceability of the procurement process.

2. Ideation Phase

2.1 Problem Statement

Laptop requests are normally raised through emails, chats or free-text forms. Requesters are unsure which configuration they are entitled to, and the procurement team has to read each request, check it against the standard and the budget, and chase missing information. The result is slow turnaround, inconsistent specifications and avoidable errors. The problem statements below were framed from the two main user perspectives.

	PS-1: Requester	PS-2: IT Procurement Coordinator
I am	an employee who needs a laptop for my job role	responsible for processing laptop orders
I'm trying to	get the right laptop quickly without going back and forth	raise accurate procurement requests for approved configurations
But	I do not know which configuration is approved for my role and my requests get sent back	requests arrive incomplete, inconsistent or outside the standard and I verify each one by hand

	PS-1: Requester	PS-2: IT Procurement Coordinator
Because	there is no single guided process and the standard specifications are not visible	requests come through emails and free-text forms with no built-in validation
Which makes me feel	frustrated and uncertain about the status	overloaded and worried about errors and inconsistent spending

2.2 Empathy Map Canvas

Two empathy maps capture what the main users say, think, do and feel today, together with their pains and the gains they expect from automation.

Empathy map 1: Employee (Requester)

SAYS "Which laptop am I allowed to ask for?" "Why was my request sent back again?"	THINKS "I just need a machine that runs my tools." "Is my request even being looked at?"
DOES - Emails IT or colleagues to find the right spec - Fills in a free-text request - Follows up for status	FEELS - Frustrated by delays - Uncertain about the status - Relieved when the laptop arrives
PAINS - No visibility of approved configurations - Repeated clarification and rework - Long, unpredictable waiting time	GAINS - A guided request based on job role - Instant recommended configuration - Clear status and notifications

Empathy map 2: IT Procurement Coordinator

SAYS "This request does not match our standard." "We need the budget and software details."	THINKS "Am I ordering the right specification?" "Too many manual checks for routine orders."
DOES - Reads each request and checks specs and budget - Chases missing details - Raises procurement requests manually	FEELS - Overloaded with repetitive work - Anxious about errors and overspend - Satisfied when orders are consistent
PAINS - Inconsistent and incomplete requests - Manual validation and data entry - Hard to spot non-standard requests	GAINS - Standard requests handled automatically - Non-standard cases flagged for decision - Consistent, traceable orders

2.3 Brainstorming

The team brainstormed ways to remove manual work from laptop ordering. Ideas were grouped, discussed and prioritised by impact on the problem and feasibility on the ServiceNow platform. Ideas that directly address the problem statements were selected for this release; the others are recorded in the future scope.

#	Idea	Description	Decision
1	Guided request by job role	Catalog form collects job role, CPU, RAM, storage, GPU, software and budget	Selected

#	Idea	Description	Decision
2	Automated recommendation	Match the request to an approved standard laptop configuration	Selected
3	Requirement and budget validation	Check inputs and budget before a request is raised	Selected
4	Automatic procurement request	Generate the procurement request without manual data entry	Selected
5	Non-standard detection and exception approval	Flag deviations and route them to an approver	Selected
6	Required software by role	Attach the required software list to the request	Selected
7	Status notifications and request report	Notify users and show basic request reporting	Selected
8	Vendor catalog integration	Pull live prices and availability from vendors	Future scope
9	Chat-based request assistant	Raise a request through a virtual agent	Future scope
10	Analytics-driven standard updates	Use order trends to refresh standard configurations	Future scope

3. Requirement Analysis

3.1 Customer Journey Map

The journey below follows a requester from raising a laptop request to receiving the device, showing where automation removes friction.

	Raise request	Validate	Recommend	Approve (if non-standard)	Procure	Track and receive
Customer action	Opens the laptop request form and enters role and needs	Submits the request	Reviews the recommended configuration	Waits for the decision on an exception	Waits for ordering	Receives updates and the laptop
Touchpoint	Service Portal catalog item	Flow Designer validation	Request confirmation	Approval notification	Procurement request	Email and portal status
Experience	Guided and quick	Instant feedback	Confident	Clear but needs patience	Hands-off	Informed
Pain point today	Unsure what to ask for	Errors found late	Spec disputes	Slow, unclear approvals	Manual re-entry	No status visibility
Automation opportunity	Role-based fields and defaults	Automatic input and budget validation	Auto-match to approved standard	Auto-detect and route exceptions	Auto-generate the procurement request	Automatic notifications

3.2 Solution Requirements

Functional requirements

ID	Requirement
FR-1	The system shall provide a laptop request form capturing employee, job role, CPU, RAM, storage, GPU, required software and budget.
FR-2	The system shall validate mandatory fields and value ranges, including budget, before processing the request.
FR-3	The system shall compare the request with the approved standard laptop configurations for the job role.
FR-4	The system shall recommend the matching approved standard configuration to the requester.
FR-5	The system shall generate a procurement request automatically for a standard request.
FR-6	The system shall detect non-standard requirements and flag the request.
FR-7	The system shall route non-standard requests to an approver and act on the approve or reject decision.
FR-8	The system shall notify the requester and approver at key stages and record every outcome.
FR-9	The system shall provide a basic report of requests by job role, type, status and processing time.

Non-functional requirements

ID	Category	Requirement
NFR-1	Performance	A recommendation shall be produced in under 3 seconds; the flow shall handle simultaneous requests without slowing down.
NFR-2	Usability	The request form shall be simple, guided and usable without training.
NFR-3	Security	Access shall be controlled by role; only approvers can decide exceptions and only administrators can change standard configurations.
NFR-4	Reliability	Flow executions shall complete without errors, with failures visible in the execution log.
NFR-5	Maintainability	Standard configurations shall be maintained as data, so they can be updated without changing the flow.
NFR-6	Auditability	Every request, decision and flow execution shall be traceable.

3.3 Data Flow Diagram

The level-1 data flow diagram shows how data moves between the users, the four main processes and the two data stores.

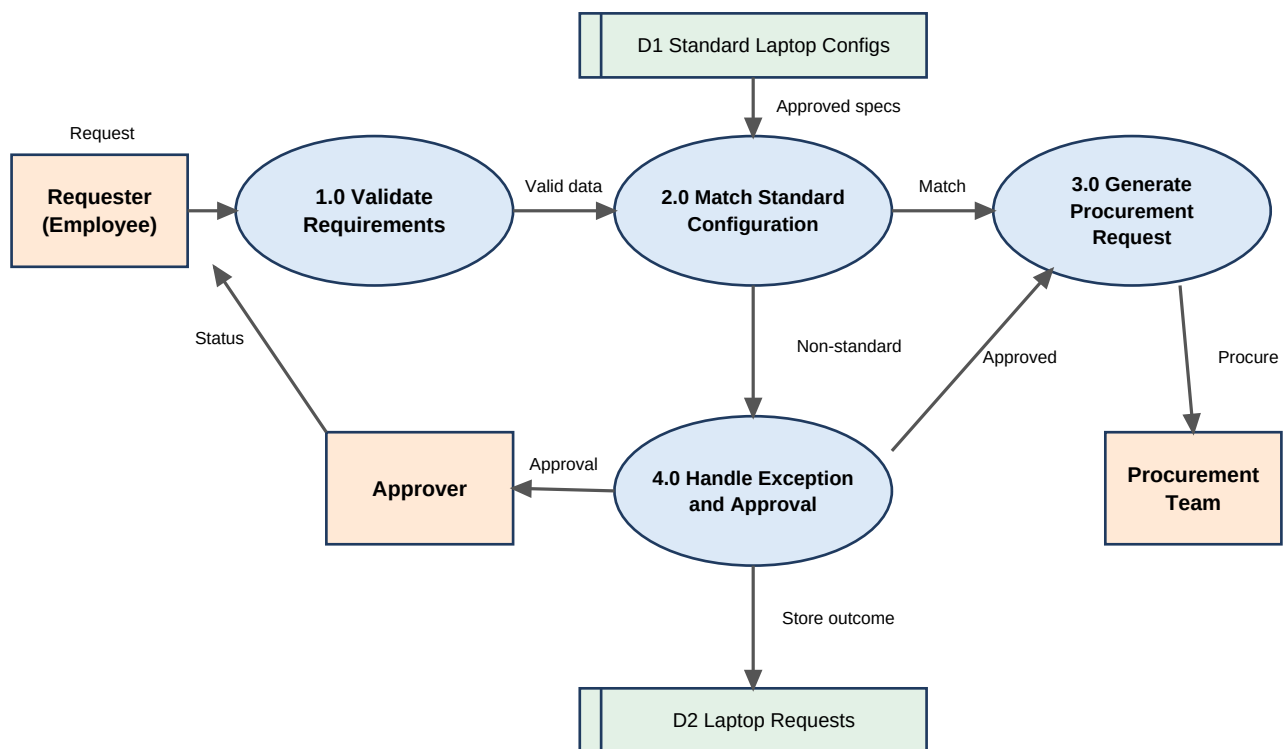


Figure 1: Level-1 Data Flow Diagram

The requester's data is validated (1.0) and matched against the standard configurations held in D1 (2.0). A standard match goes to procurement request generation (3.0). A non-standard request goes to exception handling (4.0), where the approver's decision either releases the request to 3.0 or rejects it. All outcomes are stored in D2.

3.4 Technology Stack

Layer	Technology	Purpose
Platform	ServiceNow	Hosts all components of the solution
Automation	Flow Designer (flows, subflows, actions)	Validation, matching, decision logic, approvals and request generation
Front end	Service Catalog item and Service Portal	Guided laptop request form
Data	ServiceNow tables	Standard laptop configurations, laptop requests and approval records
Approvals	Flow Designer approval action	Exception approval workflow
Notifications	ServiceNow email notifications	Status messages to requester and approver
Security	Roles and access control rules	Role-based access to requests and standards
Integration	ServiceNow REST (Table) API	Programmatic access to request data
Testing	Flow execution logs, functional and performance tests, UAT	Verification of the solution

4. Project Design

4.1 Problem Solution Fit

The table maps each user problem to the part of the solution that addresses it.

Problem	Solution fit
Requesters do not know the approved configuration for their role	Role-based guided form and an automatic recommendation
Requests are incomplete or inconsistent	Mandatory fields and automatic validation before processing
Manual checking against specs and budget	Automatic matching with the standard configurations and budget checks
Manual creation of procurement requests	Procurement request generated automatically
Non-standard requests are hard to spot	Automatic detection and routing to an approval workflow
No visibility of status	Notifications at each stage and a request report

4.2 Proposed Solution

Parameter	Description
Problem to be solved	Manual, inconsistent and slow laptop ordering that depends on emails and individual judgement.
Solution	A Flow Designer automation on ServiceNow that validates requirements, recommends an approved standard configuration, generates the procurement request and routes non-standard cases for approval.
Novelty	Treats the standard configuration as maintained data and uses it to drive validation, recommendation and exception detection in one flow, without custom code for each case.
Business impact	Less manual effort, faster turnaround, consistent specifications and a clear audit trail.
Cost efficiency	Built on the existing ServiceNow platform with low-code Flow Designer, and reduces staff time spent on routine orders.
Scalability	New job roles, configurations and approval rules can be added as data or flow steps; the same pattern can extend to other IT equipment.

4.3 Solution Architecture

The solution is arranged in four layers: users, a presentation layer (Service Portal catalog item), an automation layer (Flow Designer) and a data layer (ServiceNow tables).

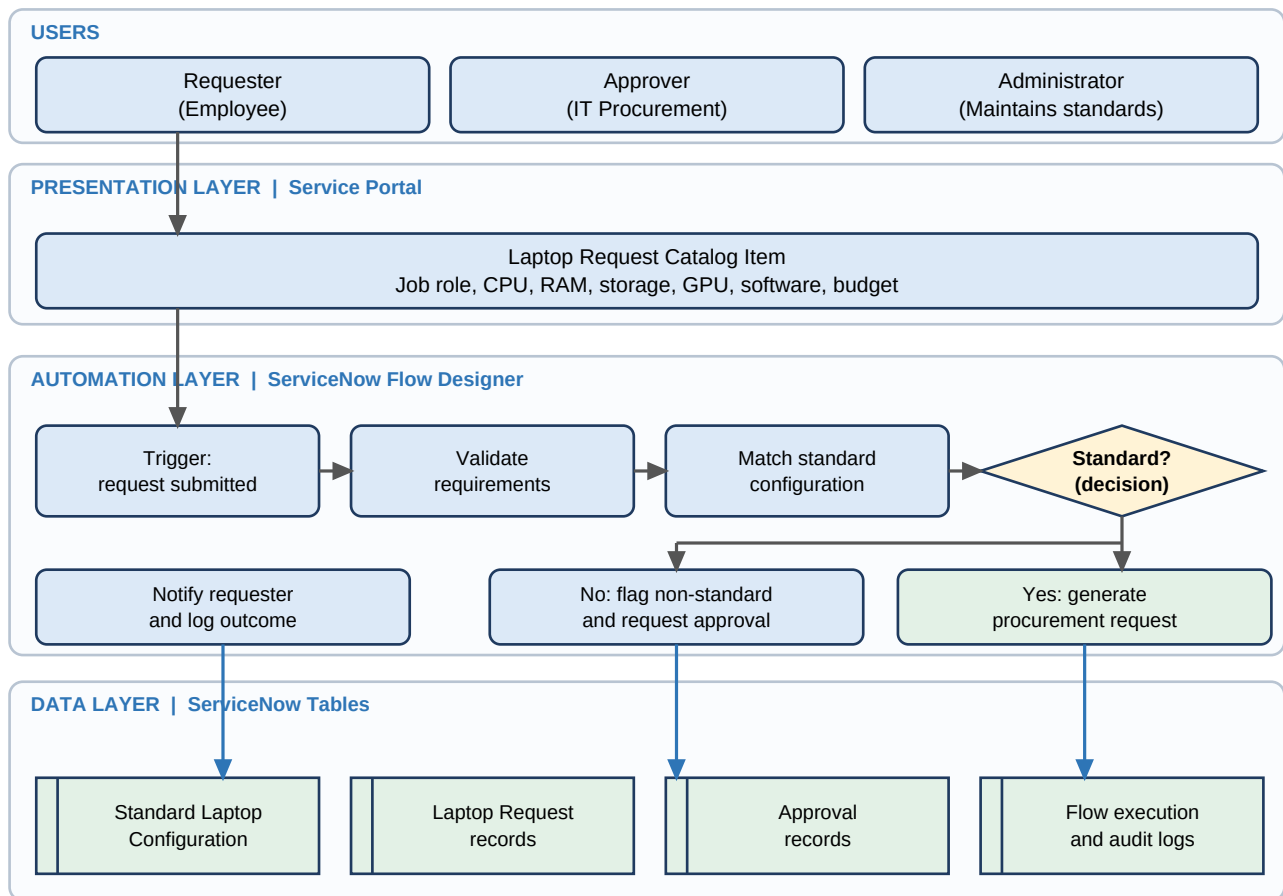


Figure 2: Solution architecture

Figure 3 shows the logic implemented by the flow, from submission to a generated procurement request or a rejection.

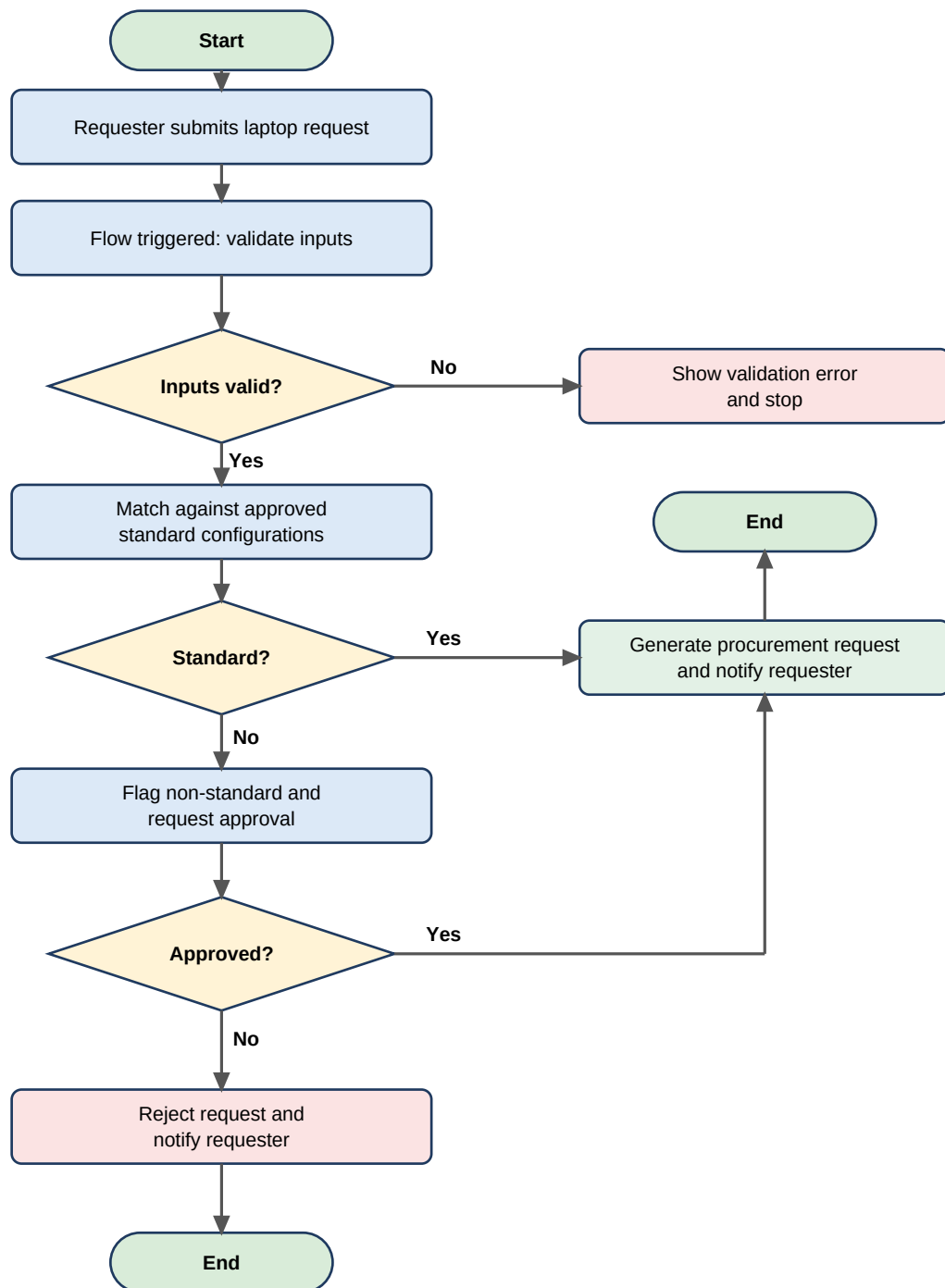


Figure 3: Laptop request automation flow

5. Project Planning and Scheduling

5.1 Project Planning

The project was planned in phases from 17 August to 02 October 2026. The schedule and the responsibilities of the team are shown below.

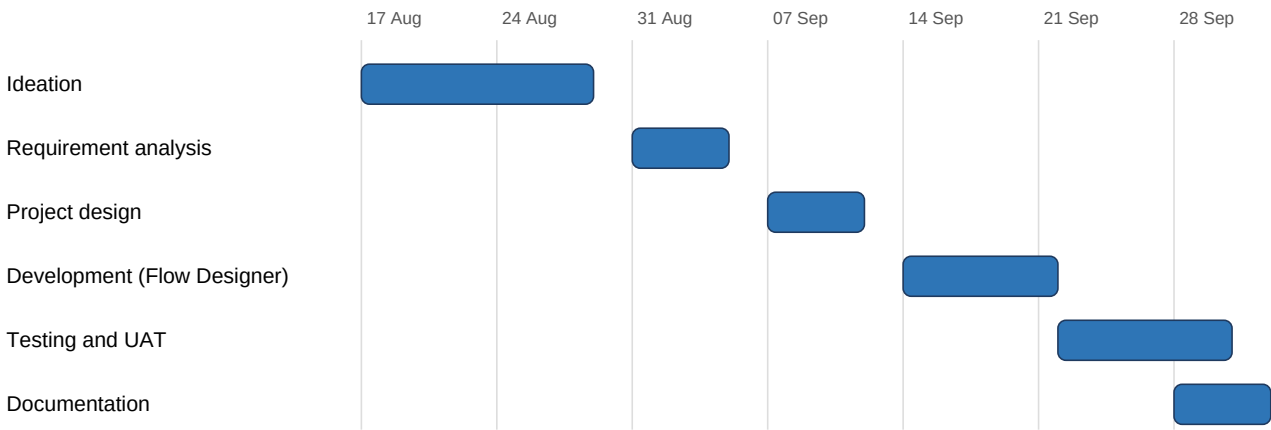


Figure 4: Project schedule

Phase	Activities	Owner	Start	End	Status
Ideation	Problem statement, empathy maps, brainstorming	Suriya Elango	17 Aug 2026	28 Aug 2026	Completed
Requirement analysis	Journey map, requirements, data flow diagram, technology stack	Suriya Elango, Rithish Nathan A	31 Aug 2026	04 Sep 2026	Completed
Project design	Problem-solution fit, proposed solution, architecture, data model	Rithish Nathan A	07 Sep 2026	11 Sep 2026	Completed
Development	Catalog item and form, flows and subflows, approval workflow	Vishal D S, Prabhash Shukla R	14 Sep 2026	21 Sep 2026	Completed
Testing and UAT	Functional, performance and user acceptance testing	Thanigaivel E	22 Sep 2026	30 Sep 2026	Completed
Documentation	Project report and technical documentation	Whole team	28 Sep 2026	02 Oct 2026	Completed

6. Functional and Performance Testing

6.1 Performance Testing

Testing covered the functional behaviour of the request flow and its performance. The test cases below were executed against the Flow Designer solution; every case met its expected result.

ID	Scenario	Expected result	Result
FT-01	Employee and job-role input validation	Valid inputs accepted; errors for invalid inputs	Pass
FT-02	Budget input validation	Valid budgets accepted; out-of-range values flagged	Pass
FT-03	Standard laptop recommendation (CPU, RAM, storage, GPU)	Matching approved configuration recommended	Pass
FT-04	Non-standard requirement detection	Request flagged and routed to approval	Pass

ID	Scenario	Expected result	Result
FT-05	Procurement request generation	Request created automatically with correct details	Pass
FT-06	Flow execution check	Flow runs end to end without errors	Pass
PT-01	Response time	Recommendation in under 3 seconds	Pass (about 2 s)
PT-02	Concurrent requests	No slowdown; all requests processed correctly	Pass
PT-03	Bulk request load	Batch processed without failures	Pass

Performance metrics

Metric	Value
Configuration match accuracy	96%
Requirement validation accuracy	98%
Non-standard detection rate	95%
Average request processing time	2 seconds
Manual effort reduction	70%

User acceptance testing was carried out from 22 to 30 September 2026. Defects found were fixed and re-tested, and all acceptance test cases passed (see the UAT report).

7. Results

7.1 Output Screenshots

The screens below show the working solution. Screenshots of the final system are inserted in the frames.

[Insert screenshot: Laptop request catalog form]

Output: Laptop request catalog form

[Insert screenshot: Recommended standard configuration]

Output: Recommended standard configuration

[Insert screenshot: Non-standard request flagged and sent for approval]

Output: Non-standard request flagged and sent for approval

[Insert screenshot: Approval decision screen]

Output: Approval decision screen

[Insert screenshot: Generated procurement request]

Output: Generated procurement request

[Insert screenshot: Flow Designer execution details]

Output: Flow Designer execution details

[Insert screenshot: Request report]

Output: Request report

Summary of results

- Standard requests were matched to an approved configuration and turned into a procurement request automatically.
- Non-standard requests were detected and routed to the approval workflow with a recorded decision.
- The recommendation time stayed within the 3-second target, and concurrent and bulk tests passed.

8. Advantages and Disadvantages

Advantages

- Reduces manual effort and shortens ordering time for standard laptops.
- Ensures that orders follow the approved configuration for the job role.
- Validates budget and requirements before a procurement request is raised.
- Detects non-standard requirements and handles them through a controlled approval.
- Provides notifications and an audit trail for every request.
- Built with low-code Flow Designer, so changes are quick to make.

Disadvantages

- The solution depends on the ServiceNow platform and its licensing.
- Accuracy depends on the standard configurations being kept up to date.
- Exception approvals still need a person to decide.
- The current scope covers standard laptops only, not other IT equipment.

9. Conclusion

The project shows that routine laptop ordering can be automated end to end with Flow Designer. The solution validates requirements, recommends an approved standard configuration, generates the procurement request and sends non-standard cases through an approval workflow. Testing confirmed that the flow works as intended and responds within the target time. The result is a faster, more consistent and more traceable procurement process that frees IT and procurement staff for work that needs their judgement.

10. Future Scope

- Integrate with vendor catalogs for live pricing and availability.
- Add a chat-based virtual agent for raising laptop requests.
- Use order and approval trends to suggest updates to the standard configurations.
- Extend the pattern to other IT equipment such as monitors, docks and mobile devices.
- Add advanced dashboards and budget forecasting for procurement managers.
- Connect fulfilment and asset management to track delivery and asset registration.

11. Appendix

Item	Details
Source code (if any)	Flow Designer flows, subflows and catalog item artifacts: [add update set or repository location]
Dataset link	Standard laptop configuration reference data: [add link]
GitHub link	[add GitHub repository link]
Project demo link	[add demo video or demo link]